

Listing Filesystem Content

Navigating the Linux filesystem can be done with a GUI or a Linux command. An IT administrator working with the shell prompt needs to become as natural as possible moving around among the different directories on your system.

The `ls` command is often used to list filesystem contents, usually for a directory.

To list files in the current directory giving a detailed listing, similar to the "properties" shown in the GUI, you can use the `ls -l` command:

```
student@ubuntu:~$ ls -l

total 204
drwxrwxr-x. 9 student student 4096 Oct 13 11:27 bashtop
drwxr-xr-x. 3 student student  20 Oct 13 11:43 Desktop
drwxr-xr-x. 2 student student   6 Oct 13 11:18 Documents
drwxr-xr-x. 2 student student   6 Oct 13 11:18 Downloads
drwxrwxr-x. 4 student student 173 Jul  6  2020 LFT
drwxr-xr-x. 2 student student   6 Oct 13 11:18 Music
drwxr-xr-x. 2 student student  24 Oct 13 13:57 Pictures
drwxr-xr-x. 2 student student   6 Oct 13 11:18 Public
-rwxr-xr-x. 1 root    root    201789 Jul  6  2020 ready-for.sh
drwxr-xr-x. 2 student student   6 Oct 13 11:18 Templates
drwxr-xr-x. 2 student student   6 Oct 13 11:18 Videos
```

To sort the list by access time, you can use the `ls -t` command:

```
student@ubuntu:~$ ls -t

file4  file2  Desktop  Downloads  Pictures  Templates
file3  file1  Documents Music      Public    Videos
```

To sort the list by access time, long listing, you can use the `ls -t -l` command:

```
student@ubuntu:~$ ls -t -l

total 32
-rw-rw-r-- 1 student student  0 Nov  2 11:50 file4
-rw-rw-r-- 1 student student  0 Nov  2 11:38 file3
-rw-rw-r-- 1 student student  0 Nov  2 11:37 file2
-rw-rw-r-- 1 student student  0 Nov  2 11:37 file1
drwxr-xr-x 2 student student 4096 Nov  2 07:25 Desktop
drwxr-xr-x 2 student student 4096 Nov  2 07:25 Documents
drwxr-xr-x 2 student student 4096 Nov  2 07:25 Downloads
drwxr-xr-x 2 student student 4096 Nov  2 07:25 Music
```

```
drwxr-xr-x 2 student student 4096 Nov 2 07:25 Pictures
drwxr-xr-x 2 student student 4096 Nov 2 07:25 Public
drwxr-xr-x 2 student student 4096 Nov 2 07:25 Templates
drwxr-xr-x 2 student student 4096 Nov 2 07:25 Videos
```

To sort the list by access time, reversed with long listing, or show the last accessed file last, you can use the `ls -lrt` command:

```
student@ubuntu:~$ ls -lrt
```

```
total 32
drwxr-xr-x 2 student student 4096 Nov 2 07:25 Videos
drwxr-xr-x 2 student student 4096 Nov 2 07:25 Templates
drwxr-xr-x 2 student student 4096 Nov 2 07:25 Public
drwxr-xr-x 2 student student 4096 Nov 2 07:25 Pictures
drwxr-xr-x 2 student student 4096 Nov 2 07:25 Music
drwxr-xr-x 2 student student 4096 Nov 2 07:25 Downloads
drwxr-xr-x 2 student student 4096 Nov 2 07:25 Documents
drwxr-xr-x 2 student student 4096 Nov 2 07:25 Desktop
-rw-rw-r-- 1 student student    0 Nov 2 11:37 file1
-rw-rw-r-- 1 student student    0 Nov 2 11:37 file2
-rw-rw-r-- 1 student student    0 Nov 2 11:38 file3
-rw-rw-r-- 1 student student    0 Nov 2 11:50 file4
```